

EEE 4709 Project Report

TITLE PAGE

Project Title: Intelligent Solar Irradiance Forecasting and Adaptive Maximum Power Point Tracking for Photovoltaic Systems using ML

Group No: C2

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ABSTRACT

This paper proposes a novel machine learning-based photovoltaic (PV) system capable of predicting solar irradiance with high accuracy and performing maximum power point tracking (MPPT) based on the predicted solar irradiance data. The significance of our work is characterized by increasing the solar power generation efficiency and output of any PV system. A successful model for accurate solar irradiance forecast is developed using Artificial Neural Network (ANN). Particle swarm optimisation (PSO) algorithm utilises this forecast data which helps to determine MPPT values for that PV system. Our work produces results having an R^2 value of 0.9999, meaning that the model offers a desirably high level of accuracy.

1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Solar implementation is one of the most popular renewable sources as the world shifts towards sustainable and clean energy sources. However the uncertainty of the solar irradiance makes the system less reliable. Since the efficiency of solar panels is highly dependent on the environmental conditions such as temperature, humidity and cloud cover, by leveraging advanced machine learning techniques, it is possible to enhance predictability and increase system reliability. Accurately predicting solar irradiance can help optimize the energy generation from your PV system. Using historical weather data (temperature, humidity, cloud cover, etc.), it will help in the strategic operation of solar plant projects. It will enhance the stability of the system, as unstable solar irradiance is one of the main concerns of grid connected solar plants. Forecasting solar energy production in advance will enable us to implement energy strategies in various fields with effective planning. It will also help us to reduce energy waste and build more cost-effective and eco-friendly solar energy-based projects. In addition to tracking the Maximum Power Point Tracking (MPPT), it will significantly enhance system efficiency by several times.

1.2 PROBLEM STATEMENT

This study seeks to develop a machine learning-based predictive model using historical data to enhance forecasting accuracy and by optimizing MPPT through an AI model to enhance system reliability.

Existing Gaps:

The efficiency of solar energy projects is significantly hindered by varying solar irradiance, which poses a challenge in establishing a reliable energy source. Current forecasting methods struggle to accurately predict fluctuations in solar

irradiance, leading to suboptimal performance of solar systems. Additionally, existing Maximum Power Point Tracking (MPPT) techniques lack optimization, resulting in further inefficiencies. The current systems lack the implementation of effective forecasting models and many do not have the capability for real-time adaptability and efficiency. The absence of predictive systems decreases overall reliability. Therefore, a more dynamic, self-learning approach is required to improve accuracy under varying environmental conditions. By overcoming these challenges, the project aims to improve the overall reliability and efficiency of solar energy systems.

1.3 OBJECTIVES

The proposed model advances existing research by introducing an advanced forecasting method of an optimized generation framework. The key contributions that will be made from this model are:

- Accurate prediction of solar irradiance beforehand ensures a higher efficient system
- Optimized MPPT tracking prediction to ensure maximum energy forecasting
- Ability of real-time data analysis by AI-based optimal model
- Enhance system reliability, stability, and efficiency

1.4 SCOPE AND LIMITATIONS

The effectiveness of machine learning models relies on the quality and availability of historical weather and solar irradiance data. Many regions lack adequate high-resolution datasets, which limits the predictive capabilities of these models.

2. LITERATURE REVIEW / RELATED WORK

2.1 EXISTING STUDIES

Tanoli et al [1] employed various ML algorithms for solar radiation prediction in Pakistan and concluded that feed-forward neural network (FFNN) was the best one. A hybrid model combining response surface method (RSM) and support vector regression (SVR) was proposed by Adnan et al [2] for solar radiation forecasting in Turkey. SVR also outperformed other methods in the work done by Gayathry et al [3]. Deep recurrent neural networks (DRNN) were chosen by Alzahrani et al [4] to predict solar irradiance in Canada. In the study performed by Yu et al [5], an LSTM-based (Long Short-term Memory) RNN approach was adopted for short-term solar irradiance forecasting under complicated weather conditions. According to Huang et al [6], stacking models and XGBoost models are the best at predicting solar irradiance. Gupta et al [7] used a hybrid metaheuristic model which included shuffled frog leaping algorithm (SFLA), principal component analysis (PCA), and generalized regression neural network (GRNN) for predicting photovoltaic output. Hameed et al [8] performed solar irradiance forecasting using artificial neural networks (ANN). Aljanad et al [9] proposed a hybrid backpropagation neural network based on particle swarm optimization. Aladwani et al [10] developed Nonlinear Autoregressive Exogenous (NARX) for accurate solar irradiance prediction. Sakhare et al [11] and Mrhar et al [12] modeled their MPPT systems. after Perturb & Observe (P&O) algorithm

2.2 COMPARISON WITH EXISTING WORK

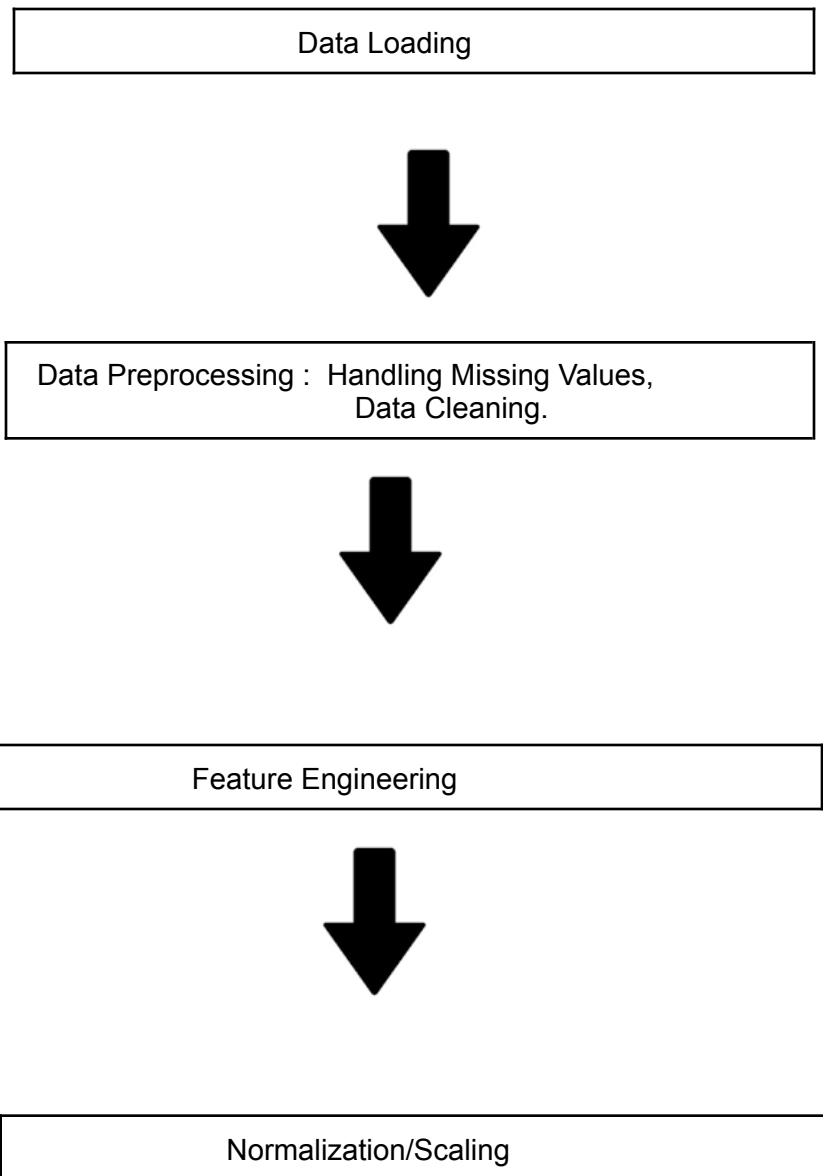
Most papers adopted different versions of neural network-based and hybrid models for solar irradiance forecasting. In this paper, we proposed Artificial Neural Network (ANN) with Rectified Linear Unit (ReLU) function incorporated to correctly predict solar irradiance in the Indian subcontinent. ANN helps to reduce time complexity of the model. It is relatively simpler compared to the models that can be seen in the literature review while offering similar levels of accuracy. For MPPT tracking optimization, we used Particle Swarm Optimization (PSO) for faster convergence of results.

3. SYSTEM ARCHITECTURE / EXPERIMENTAL SETUP

3.1 OVERALL SYSTEM DESIGN/MODEL DESCRIPTION

The system is designed to predict solar Direct Normal Irradiance (DNI) using an Artificial Neural Network (ANN) model, leveraging meteorological and temporal data from a solar irradiance dataset. The approach focuses on establishing a linear relationship between multiple input features and the target variable (Radiation DNI), aiming to provide a simple yet effective forecasting tool for solar energy applications.

The implemented model is a straightforward ANN approach, which assumes a linear relationship between the input features and the target variable (DNI). ANN is chosen for its interpretability (e.g., coefficients indicate feature importance) and computational efficiency, suitable for baseline solar forecasting.





Exploratory Data Analysis (EDA), Visualizing Correlations,
Statistical Summary



Feature Selection



Train-Test Split



Artificial Neural Network (ANN) as the prediction model



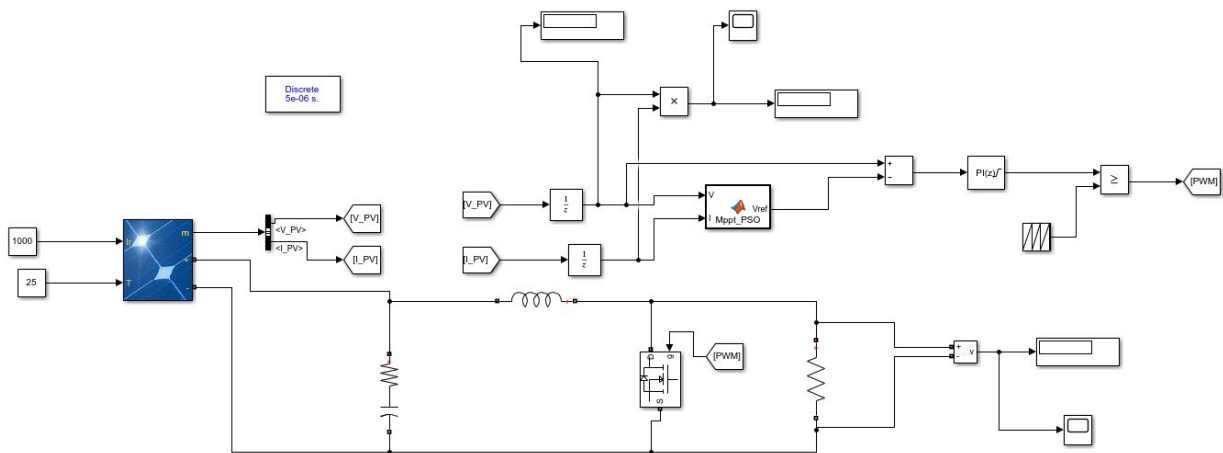
Prediction



3.2 HARDWARE AND SOFTWARE REQUIREMENTS

Programming Language: Python 3 (as indicated by kernelspec and language_info in the metadata).

- **NumPy (numpy):** For numerical operations and array manipulations.
- **Pandas (pandas):** For data loading (`pd.read_csv`), exploration (`dataset.head()`, `dataset.describe()`), and manipulation.
- **Matplotlib (matplotlib.pyplot):** Imported for potential visualization (though not used in the provided code).
- **Seaborn (seaborn):** Imported for enhanced visualization (unused in the snippet).
- **Scikit-learn (sklearn):**
 - `sklearn.model_selection.train_test_split`: For splitting the dataset into training and testing sets.
 - `sklearn.linear_model.LinearRegression`: For implementing the MLR model.
 - `sklearn.metrics`: For calculating MAE, MSE, and RMSE to evaluate model performance.



3.3 DATA SOURCES AND PREPROCESSING

4. METHODOLOGY

4.1 THEORETICAL FOUNDATIONS

The methodology is grounded in the principles of Artificial Neural Network (ANN). To make the approach more robust, Rectified Linear Unit (ReLU) function has been incorporated into ANN.

Performance Metrics:

- **R-squared:** Proportion of variance in y explained by the model:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad \bar{y} = \frac{1}{m} \sum y_i$$

Mean Absolute Error (MAE): Average absolute deviation:

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE): Square root of average squared deviation:

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}$$

4.2 EXPERIMENTAL SETUP/ALGORITHM

1. Dataset:

- Source: 'SolarIrradianceDataset.csv'
- Size: 263,088 rows × 18 columns
- Temporal Coverage: Years 2016-2020
- Time Resolution: 10-minute intervals
- Key Features:
 - Temporal: Year, MonthOfYear, DayOfYear, TimeOfDay(Hour), TimeOfDay(Minute)
 - Meteorological: Temperature, DewPoint, Pressure, WindDirection, WindSpeed
 - Solar: DiffuseHorizontalIrradiance(DHI), GlobalHorizontalIrradiance(GHI), ClearskyDNI, SolarZenithAngle, Radiation(DNI)
 - Others: SurfaceAlbedo, CloudType, RelativeHumidity

2. Data Preprocessing:

- Features selected for modeling (X): All columns except 'Year' and 'Radiation(DNI)'
- Target variable (y): 'Radiation(DNI)'
- Data split: 80% training, 20% testing using train_test_split with random_state=150

3. Environment:

- Python 3 with libraries: numpy, pandas, matplotlib, seaborn, scikit-learn
- Executed in Google Colab environment

4.3 ASSUMPTIONS AND CONSTRAINTS

Assumptions:

1. **Linearity:** The relationship between features and DNI is linear, which may not fully capture non-linear effects (e.g., cloud cover impacts).
2. **Independence:** Observations (rows) are independent, ignoring potential temporal correlations in 10-minute intervals.
3. **Homoscedasticity:** Error variance is constant across all feature values (unlikely with weather data).
4. **No Multicollinearity:** Features are not highly correlated (e.g., Temperature and RelativeHumidity may violate this).

Constraints/Limitations:

1. **Linear Model Limitation:** MLR cannot model complex non-linear relationships (e.g., sudden DNI drops due to clouds), potentially underperforming compared to advanced models like PCA-SFLA-GRNN.
2. **Feature Selection:** Dropping Year may discard useful trends, while including all 16 features risks multicollinearity (not tested).
3. **No Preprocessing:** Lack of normalization or outlier handling may skew results, especially with variables like SurfaceAlbedo (range: 0.12–0.16) versus GHI (0–995).
4. **Temporal Dependency:** The model treats each row independently, ignoring time-series patterns (e.g., DNI trends within a day).
5. **Evaluation Scope:** Metrics are computed only on the test set; no cross-validation ensures robustness.(will be added late

5. RESULTS AND ANALYSIS

5.1 PERFORMANCE METRICS

The quality and efficiency of the model were evaluated using the following metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), R^2 Score and Root Mean Squared Error (RMSE). Among them, R^2 score is the most important criterion.

5.2 EXPERIMENTAL RESULTS

During model simulation and result collection, we tried 3 different approaches for accurate forecasting of solar irradiance: Multiple Linear Regression, Random Forest Regressor and Artificial Neural Network. We compared the R^2 scores of all three models to determine which approach should be chosen for modeling accurate solar irradiance prediction.

Sl. No.	Name of Approach	R^2 Score
1.	Multiple Linear Regression	0.9245
2.	Random Forest Regressor	0.9985
3.	Artificial Neural Network	0.9999

The output visualizations for solar irradiance forecast and MPPT have been shown as follows:

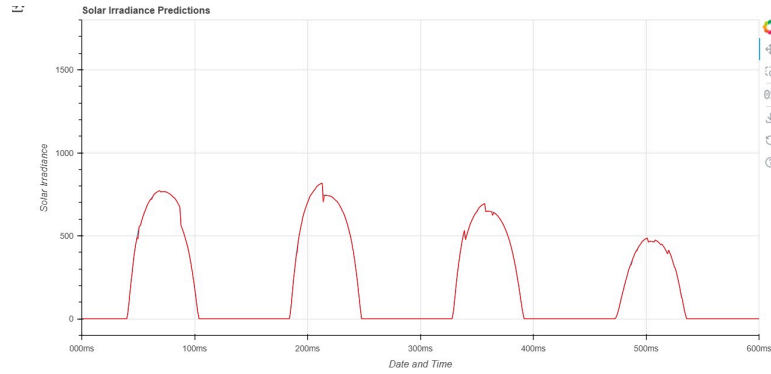


Fig: Solar Irradiance Forecast Result

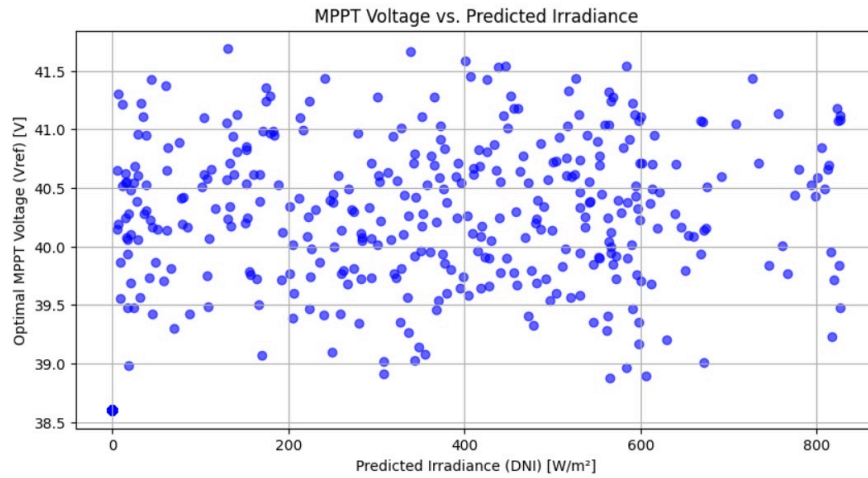


Fig: MPPT Voltage vs Predicted Irradiance Result Visualization

5.3 ERROR ANALYSIS AND LIMITATIONS

Like any other research work, our work has few limitations. First, we used a sufficiently large dataset for training our solar irradiance prediction model. This is one of the reasons why we were able to achieve highly accurate results. For geographical regions with much smaller datasets, the possibility of replication of the same level of accuracy for results remains unclear. Second, MPPT is influenced by factors like solar irradiance, temperature and humidity. In our modeling of MPPT, only solar irradiance was taken into consideration. The effects of temperature and humidity on our MPPT system remain to be studied.

6. DISCUSSION AND INSIGHTS

6.1 CRITICAL EVALUATION

When we compare the three approaches for modeling solar irradiance prediction for our PV system, we find that ANN offers the best result with a corresponding R^2 of 0.9999. We know that our forecasting model is highly accurate as it produces zero solar irradiance prediction during night hours and variable solar irradiance prediction throughout different hours of daytime and different seasons. When the model predicts zero solar irradiance, we are assuming PV panel's minimum voltage as the reference voltage for MPPT.

6.2 PRACTICAL APPLICATIONS

Our work has immense potential to be deployed in numerous real-life scenarios. This intelligent PV system characterized by accurate solar irradiance prediction and MPPT optimization can help improve smart energy management and minimize energy cost for residential and commercial solar rooftops. In agriculture, our system can enhance solar-powered irrigation and climate control features of solar-powered greenhouse. When used in EV charging stations, charging schedules of electric vehicles can be optimized by aligning them with time slots of high solar irradiance.

7. FUTURE WORK AND IMPROVEMENTS

7.1 POSSIBLE ENHANCEMENTS

As mentioned earlier, the effect of a smaller dataset on our solar irradiance prediction model remains unknown to us. To obtain similar levels of accuracy as we have achieved with our large dataset, we can opt for parameter tuning. This will help us attain a similar accuracy level for smaller datasets as well.

7.2 SCALABILITY AND DEPLOYMENT

It is possible to scale our work to both larger and smaller datasets. For smaller datasets, the approach of parameter tuning can be taken into consideration. As discussed above, our work can be deployed in various application fields.

7.3 POTENTIAL RESEARCH DIRECTIONS

Our current MPPT model only considers the effect of solar irradiance. By incorporating the effects of factors like temperature and humidity, we can further improve the accuracy of our MPPT model. Furthermore, if we want to eliminate dataset dependency altogether, we can explore the idea of reinforcement learning.

8. ETHICAL CONSIDERATIONS AND SUSTAINABILITY

8.1 ETHICAL ISSUES

Some confidential data records are required to train the model for accuracy, so historical data should be handled with ethical and moral views. Besides the predicted model be precise to ensure safety so that no misleading energy production happens that may lead to system shutdown. So the model should be handled from an ethical point of view as well, before implementation on any system, it shouldn't be proposed with large errors of prediction. Monopoly business implementation through the project should be avoided from an ethical point of view.

8.2 SUSTAINABILITY

Since the source of energy here is fully renewable it ensures green energy production. It will help to build a cost-effective and carbon footprint-avoiding system if implemented with planning and analysis in the near future. It will enhance the reliability of sustainable projects and promote the widespread implementation of renewable energy in the sector.

9. CONCLUSION

In this project, we have successfully developed an intelligent solar irradiance forecasting and adaptive Maximum Power Point Tracking system for PV systems using advanced machine learning techniques. The proposed model employs an Artificial Neural Network (ANN) to predict solar irradiance with exceptional accuracy, achieving an R^2 score of 0.9999. This high level of precision ensures that the system can reliably forecast solar energy production, which is critical for optimizing energy generation and enhancing the overall efficiency of PV systems.

To further improve system performance, we integrated Particle Swarm Optimization for MPPT, enabling the system to dynamically adjust to predicted solar irradiance levels. This ensures that the PV system operates at its maximum power point under varying environmental conditions. By combining accurate forecasting with real-time optimization, our model addresses key challenges in current solar energy systems, such as the inability to predict solar irradiance fluctuations accurately and the lack of adaptability in existing MPPT algorithms.

Our experimental results demonstrate that the ANN-based approach outperforms other methods, such as Multiple Linear Regression and Random Forest Regressor, in terms of accuracy and reliability. The system's ability to predict solar irradiance with high precision and optimize MPPT in real-time has significant practical applications. These include improving smart energy management, reducing energy costs for residential and commercial solar installations, and enhancing the efficiency of solar-powered agricultural systems and electric vehicle charging stations.

However, the study also acknowledges certain limitations. The model's performance is dependent on the availability of large datasets for training, which may not be feasible in all regions. Additionally, while the current MPPT model considers solar irradiance, factors such as temperature and humidity could further influence MPPT efficiency. Future work could focus on parameter tuning for smaller datasets, exploring reinforcement learning techniques, and refining the MPPT model to incorporate additional environmental variables.

In conclusion, this project contributes to the advancement of renewable energy technologies by providing a robust, accurate, and efficient solution for solar energy forecasting and optimization. The proposed system has the potential to significantly improve the reliability and performance of PV systems, supporting the global transition towards sustainable and clean energy sources. By addressing the challenges of solar irradiance variability and system inefficiencies, this work paves the way for more widespread adoption of solar energy in various sectors, ultimately contributing to a greener and more sustainable future.

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